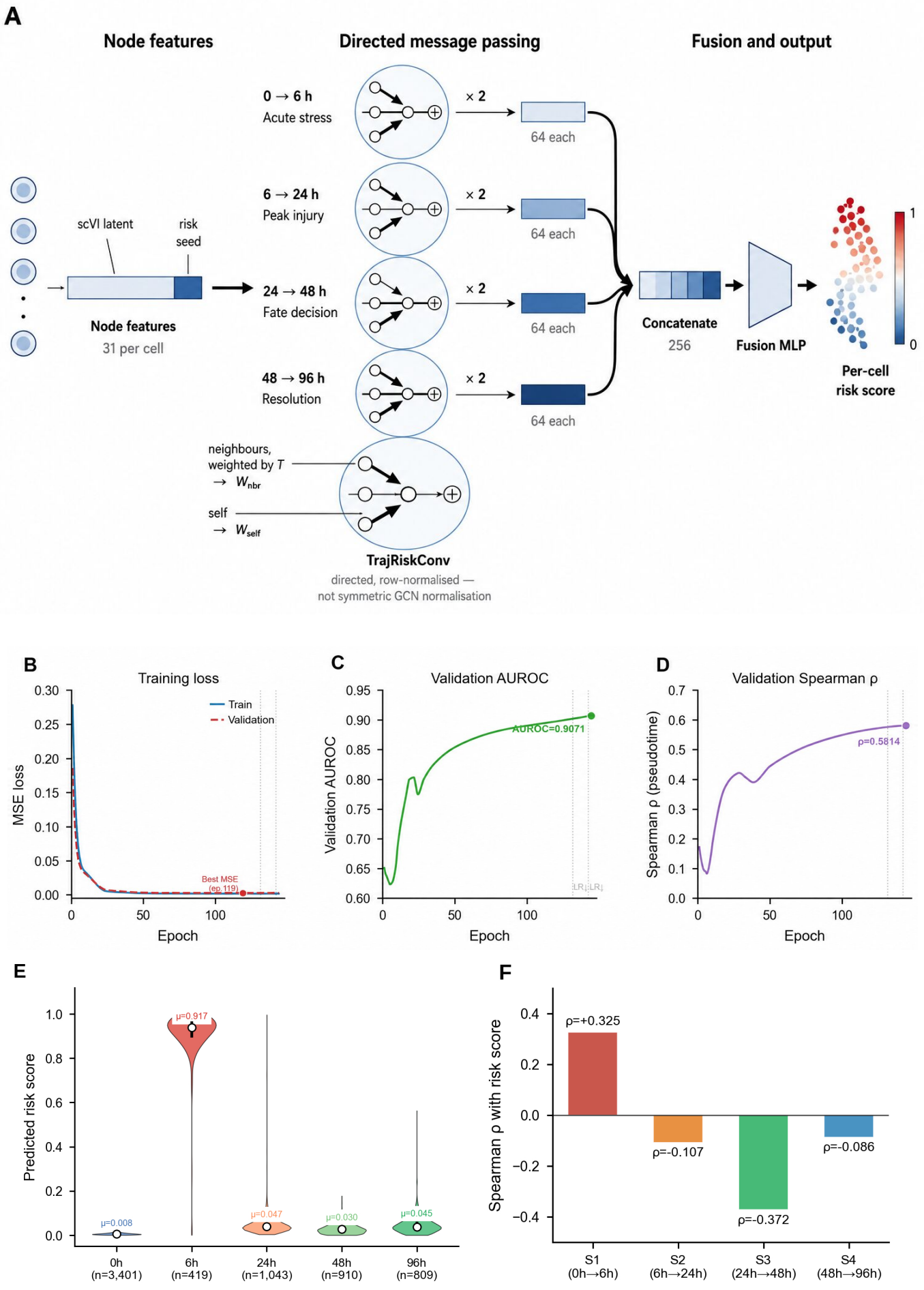

A
Node features
Directed message passing
Fusion and output
scVI latent
risk seed
Node features
31 per cell
0 → 6 h
Acute stress
6 → 24 h
Peak injury
24 → 48 h
Fate decision
48 → 96 h
Resolution
× 2
64 each
neighbours, weighted by T
→ W_nbr
self
→ W_self
TrajRiskConv
directed, row-normalised — not symmetric GCN normalisation
Concatenate
256
Fusion MLP
Per-cell risk score
1
0
B
Training loss
Train
Validation
MSE loss
Best MSE (ep 119)
Epoch
C
Validation AUROC
Validation AUROC
AUROC=0.9071
Epoch
D
Validation Spearman ρ
Spearman ρ (pseudotime)
ρ=0.5814
Epoch
E
Predicted risk score
μ=0.008
μ=0.917
μ=0.047
μ=0.030
μ=0.045
0h (n=3,401)
6h (n=419)
24h (n=1,043)
48h (n=910)
96h (n=809)
F
Spearman ρ with risk score
ρ=+0.325
ρ=-0.107
ρ=-0.372
ρ=-0.086
S1 (0h→6h)
S2 (6h→24h)
S3 (24h→48h)
S4 (48h→96h)